Problem 2

Part (a)

An integer $n \in \mathbb{N}$ satisfies $m \mid n$ if and only if $\exists k \in \mathbb{N} : n = mk$. Therefore:

$$A_m = \bigcup_{k \in \mathbb{N}} \{mk\}$$

Since the sets $\{mk\}_{k \in \mathbb{N}}$ are pairwise disjoint, we can apply σ-additivity:

$$\begin{aligned}
\mathbb{P}_s(A_m) &= \mathbb{P}_s\left(\bigcup_{k \in \mathbb{N}} \{mk\}\right) = \sum_{k=1}^{\infty} \mathbb{P}_s(\{mk\}) && \left(\mathbb{P}_s(\{n\}) = \frac{1}{\zeta(s) n^s}\right) \\
&= \sum_{k=1}^{\infty} \frac{1}{\zeta(s)(mk)^s} = \frac{1}{\zeta(s) m^s} \sum_{k=1}^{\infty} \frac{1}{k^s} && \left(\zeta(s) = \sum_{k=1}^{\infty} \frac{1}{k^s}\right) \\
&= \frac{1}{\zeta(s) m^s} \cdot \zeta(s) = \frac{1}{m^s}
\end{aligned}$$

Therefore, $\mathbb{P}_s(A_m) = \frac{1}{m^s}$.

$\square$

Part (b)

Let $J = \{k_1, k_2, \ldots, k_n\} \subset P$. We must show that $\mathbb{P}_s\left(\bigcap_{i=1}^n A_{k_i}\right) = \prod_{i=1}^n \mathbb{P}_s(A_{k_i})$.

For any $x \in \mathbb{N}$, we have:

$$\begin{aligned}
x \in \bigcap_{i=1}^n A_{k_i} &\iff x \in A_{k_i} \quad \forall i \in \{1, \ldots, n\} \\
&\iff k_i \mid x \quad \forall i \in \{1, \ldots, n\}
\end{aligned}$$

Because $k_1, \ldots, k_n$ are distinct prime numbers, they are pairwise coprime. Therefore, if they all divide x, their product must also divide x:

$$\begin{aligned}
k_i \mid x \quad \forall i \in \{1, \ldots, n\} &\iff \left(\prod_{i=1}^n k_i\right) \Bigg| \, x \\
&\iff x \in A_{\prod_{i=1}^n k_i}
\end{aligned}$$

This chain of equivalences for all elements x establishes the set equality:

$$\bigcap_{i=1}^n A_{k_i} = A_{\prod_{i=1}^n k_i}$$

Applying the probability measure $\mathbb{P}_s$ to both sides, and using the result from part (a) where $\mathbb{P}_s(A_m) = \frac{1}{m^s}$:

$$\begin{aligned}
\mathbb{P}_s\left(\bigcap_{i=1}^n A_{k_i}\right) &= \mathbb{P}_s\left(A_{\prod_{i=1}^n k_i}\right) \\
&\overset{\text{(part a)}}{=} \frac{1}{\left(\prod_{i=1}^n k_i\right)^s} \\
&= \prod_{i=1}^n \frac{1}{k_i^s} \\
&= \prod_{i=1}^n \mathbb{P}_s(A_{k_i})
\end{aligned}$$

Since $J \subset P$ was an arbitrarily chosen finite subset, the product rule holds for all finite subsets of P. Therefore, the events $(A_p)_{p\in P}$ are independent. $\square$

Part (c)

The event A_p^c contains all natural numbers not divisible by p. Thus, the intersection $\bigcap_{p\in P} A_p^c$ is the set of all natural numbers not divisible by *any* prime. The only natural number without any prime factors is 1 (by definition, P excludes 1). Therefore:

$$\bigcap_{p\in P} A_p^c = \{1\}$$

By the definition of $\mathbb{P}_s$:

$$\mathbb{P}_s\left(\bigcap_{p\in P} A_p^c\right) = \mathbb{P}_s(\{1\}) = \frac{1}{\zeta(s)\cdot 1^s} = \frac{1}{\zeta(s)} \tag{1}$$

Enumerate the primes in increasing order: $P = \{p_1, p_2, \dots\}$. We have that the sets in $(A_{p_i})_{i\in\mathbb{N}}$ are independent and, to evaluate the probability of the intersection of complements $(A_{p_i}^c)_{i\in\mathbb{N}}$, we first establish their independence. Let $\mathcal{M}_i = \{A_{p_i}\}$ for $i \in \mathbb{N}$. Because the events $(A_{p_i})_{i\in\mathbb{N}}$ are independent, the family of intersection-stable generators $(\mathcal{M}_i)_{i\in\mathbb{N}}$ is independent (Definition 2.2.9). Consequently, the family of generated σ-algebras $\sigma(\mathcal{M}_i) = \{\emptyset, A_{p_i}, A_{p_i}^c, \mathbb{N}\}$ is independent (Lemma 2.2.11). Since $A_{p_i}^c \in \sigma(\mathcal{M}_i)$ for all $i \in \mathbb{N}$, the sequence of complements $(A_{p_i}^c)_{i\in\mathbb{N}}$ is also independent (Definition 2.2.9). Then, by independence:

$$\begin{aligned}
\mathbb{P}_s\left(\bigcap_{p\in P} A_p^c\right) &= \mathbb{P}_s\left(\bigcap_{i\in\mathbb{N}} A_{p_i}^c\right) \\
&\overset{\text{(Lemma 2.2.5)}}{=} \prod_{i\in\mathbb{N}} \mathbb{P}_s(A_{p_i}^c) \\
&= \prod_{p\in P} (1 - \mathbb{P}_s(A_p)) \\
&\overset{\text{(part a)}}{=} \prod_{p\in P}\left(1 - \frac{1}{p^s}\right)
\end{aligned} \tag{2}$$

Equating (1) and (2), we arrive at Euler's formula :

$$\frac{1}{\zeta(s)} = \prod_{p\in P}\left(1 - \frac{1}{p^s}\right)$$

Part (d)

First, we prove the set equality $\bigcap_{m\in\mathbb{N}\setminus\{1\}} A_{m^2}^c = \bigcap_{p\in P} A_{p^2}^c$, by double inclusion:

- "$\subseteq$": Let $x \in \bigcap_{m\in\mathbb{N}\setminus\{1\}} A_{m^2}^c$, i.e., $\nexists m \geq 2 : m^2 \mid x$. Since $P \subset \mathbb{N}\setminus\{1\}$, then $\nexists p \in P : p^2 \mid x$. Therefore, $x \in \bigcap_{p\in P} A_{p^2}^c$, i.e., $\bigcap_{m\in\mathbb{N}\setminus\{1\}} A_{m^2}^c \subseteq \bigcap_{p\in P} A_{p^2}^c$.
- "$\supseteq$": Let $x \in \bigcap_{p\in P} A_{p^2}^c$, i.e., $\nexists p \in P : p^2 \mid x$. Let $m \geq 2$ be an arbitrary natural number. Then:

$$\begin{aligned}
& \exists p \in P : p \mid m \\
\implies & \exists k \in \mathbb{N} : m = kp \\
\implies & \exists k \in \mathbb{N} : m^2 = k^2 p^2
\end{aligned}$$

Assume by contradiction that $m^2 \mid x$. Then:

$$m^2 \mid x \implies k^2 p^2 \mid x \implies p^2 \mid x$$

which contradicts $\nexists p \in P : p^2 \mid x$. Therefore, $\nexists m \geq 2 : m^2 \mid x$. Since m was arbitrarily fixed, $x \in \bigcap_{m \in \mathbb{N} \setminus \{1\}} A_{m^2}^c$ and thus $\bigcap_{p \in P} A_{p^2}^c \subseteq \bigcap_{m \in \mathbb{N} \setminus \{1\}} A_{m^2}^c$.

Because the sets are identical, their probabilities are equal: $\mathbb{P}_s \left(\bigcap_{m \in \mathbb{N} \setminus \{1\}} A_{m^2}^c \right) = \mathbb{P}_s \left(\bigcap_{p \in P} A_{p^2}^c \right)$.

Second, we show the independence of the family $(A_{p^2})_{p \in P}$. Let $J = \{p_1, p_2, \dots, p_n\} \subset P$ be an arbitrary finite subset of distinct primes. Because distinct primes are pairwise coprime, their squares $p_1^2, \dots, p_n^2$ are also pairwise coprime. Therefore, by the exact same divisibility logic established in part (b), the intersection of their events is equivalent to the event of their product:

$$\bigcap_{i=1}^{n} A_{p_i^2} = A_{\prod_{i=1}^{n} p_i^2}$$

Applying the probability measure $\mathbb{P}_s(A_m) = \frac{1}{m^s}$ from part (a) to this set equality yields the product rule directly:

$$\begin{aligned}
\mathbb{P}_s \left(\bigcap_{i=1}^{n} A_{p_i^2} \right) &= \mathbb{P}_s \left(A_{\prod_{i=1}^{n} p_i^2} \right) \\
&= \frac{1}{\left(\prod_{i=1}^{n} p_i^2 \right)^s} \\
&= \prod_{i=1}^{n} \frac{1}{(p_i^2)^s} \\
&= \prod_{i=1}^{n} \mathbb{P}_s(A_{p_i^2})
\end{aligned}$$

Since J was an arbitrarily chosen finite subset, the product rule holds for all finite subsets of P. Therefore, the family of events $(A_{p^2})_{p \in P}$ is stochastically independent.

Third, we evaluate the probability using the independent family of events $(A_{p^2})_{p \in P}$. By enumerating the primes $P = \{p_1, p_2, \dots\}$ and applying Lemma 2.2.5:

$$\begin{aligned}
\mathbb{P}_s \left(\bigcap_{m \in \mathbb{N} \setminus \{1\}} A_{m^2}^c \right) &= \mathbb{P}_s \left(\bigcap_{i=1}^{\infty} A_{p_i^2}^c \right) \\
&= \prod_{i=1}^{\infty} \mathbb{P}_s(A_{p_i^2}^c) && \text{(Lemma 2.2.5)} \\
&= \prod_{p \in P} (1 - \mathbb{P}_s(A_{p^2})) && \text{(part a)} \\
&= \prod_{p \in P} \left(1 - \frac{1}{p^{2s}} \right)
\end{aligned}$$

Since $s > 1$ implies $2s > 1$, Euler's formula from part (c) applies with $2s$ in place of s, giving $\prod_{p \in P} \left(1 - \frac{1}{p^{2s}} \right) = \frac{1}{\zeta(2s)}$. Therefore:

$$\mathbb{P}_s \left(\bigcap_{m \in \mathbb{N} \setminus \{1\}} A_{m^2}^c \right) = \frac{1}{\zeta(2s)}$$

□